Zero-Knowledge Zakat: Privacy, Transparency, and Sharia Compliance on the Blockchain

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***Abstract*—Zakat, the annual wealth redistribution that every Muslim above the nisab threshold owes, has moved onto blockchain. Several platforms exist now, and every single one logs complete donor-recipient histories on public ledgers for anyone to read. Who gave, how much, and to whom. All permanently visible. That visibility collides with *nafs*, the principle of dignity that maqasid al-shariah ranks among its five essential protections. On Ethereum Sepolia, we built a prototype that breaks that tradeoff. Donors submit zakat through UltraHONK zero-knowledge proofs. A Noir circuit (29 ACIR opcodes) verifies nisab and hawl eligibility without exposing private inputs, a Pedersen-hash nullifier stops double-claims, and a four-tier model gives donors control over what they reveal. Benchmarking five runs on Barretenberg v4.2.1 across 16 Linux cores, proof generation averaged 246.8 ms. One end-to-end `donateZK()` call on Sepolia consumed 721,345 gas. What emerges is that privacy and accountability need not oppose each other. Amil institutions get verifiable aggregate records while individual amounts and identities stay hidden. The protocol brings a privacy layer to Islamic social finance, grounded in nafs protection.**

***Index Terms*—Barretenberg, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Pedersen Hash, Privacy-Preserving Donations, Sharia Compliance, Soulbound Token, Zakat, Zero-Knowledge Proofs**

I. Introduction

Blockchain zakat management faces practical hurdles. BAZNAS [1] estimates Indonesia's annual zakat potential at Rp 327 trillion (approximately USD 20 billion). Actual collection remains far lower.

Earlier work chases speed and traceability. A decentralized collection and distribution platform was demonstrated by Khan [2]. Nazeri et al. [3] landed on a conclusion that, from our view, exposes the problem. Its premise, that blockchain's immutable ledger gives complete traceability of zakat funds from donor to asnaf, is what this paper challenges. Khairi et al. [4] took the idea further, adding automated disbursement and an immutable audit trail on ERC-20 tokens. Every transaction, donor address and amount and recipient pool, sits permanently on a public ledger. For recipients, most of whom fall below the nisab threshold, that audit trail broadcasts socioeconomic status indefinitely.

Maqasid al-shariah addresses this directly. Al-Shatibi's *al-Muwafaqat* classifies *hifz al-nafs* (preservation of life) among the five *daruriyyat* (necessities) [5]. Ibn Ashur's *Maqasid al-Shariah al-Islamiyyah* stretches this to cover *al-karamah*. Exposing financial weakness, he argues, violates the maqsad of protecting the self [6]. Classical sources treat concealed charity (*sadaqah al-sirr*) as superior to public giving for exactly this reason. It guards the recipient's honor. A blockchain ledger that permanently records zakat eligibility on Etherscan collides with the same principle.

Zero-knowledge proofs offer a path through. Prove eligibility cryptographically. Do not reveal the underlying data. Hameed et al. [7] and Wen et al. [8] have surveyed the ZKP landscape. Noir [10] compiles to an intermediate representation, ACIR, which Barretenberg's UltraHONK prover [9] turns into a succinct proof, verifiable on-chain, the witness hidden. Separately, Buterin et al. [15] argue that ZKPs can reconcile privacy with regulatory goals.

We combine these pieces into a zakat-specific protocol. The Noir circuit (`main.nr`) encodes two Sharia predicates. Total wealth must fall below the nisab threshold of 85,000,000 IDR. Assets must be held for at least one lunar year (hawl). A Pedersen-hash nullifier, $\text{pedersen}(secret, recipient\_address, cycle\_id)$, stops the same recipient from claiming twice per cycle, with no identity exposure needed. Proving runs client-side. The pipeline is `nargo compile` $\rightarrow$ `nargo execute` $\rightarrow$ `bb prove`. The proof and nullifier go to `ZKTCore.donateZK()` on Ethereum Sepolia for settlement. Two research questions follow. Can off-chain Noir proving plus on-chain nullifier tracking serve institutional zakat accountability, and does the resulting protocol satisfy nafs protection under maqasid al-shariah.

The paper makes several contributions. A Noir circuit encodes nisab and hawl as verifiable predicates. Its assertion `total_wealth < nisab_threshold` directly instantiates the Sharia requirement that zakat recipients fall below the wealth threshold. A Pedersen-hash nullifier mechanism blocks re-claiming without identity exposure, enforced on-chain by `NullifierRegistry.spend()` rejecting duplicates. A four-tier donor privacy model (Private, Semi-Private, Verified-Private, Public) gives donors control over disclosure. The Sepolia prototype generates UltraHONK proofs in 246.8 ms on average ($\sigma = 9.3$ ms across five runs) and completes end-to-end donations in 721,345 gas.

II. Literature Review

Blockchain-based zakat management has drawn research attention. Khan's decentralized platform [2] set the baseline, handling collection and distribution without intermediaries. In Malaysia, Nazeri et al. [3] explored how blockchain features align with zakat objectives. Khairi et al. [4] developed a full system with ERC-20 tokens providing automated disbursement and an immutable audit trail. Blockchain's potential and challenges for zakat institutions were examined in a recent Emerald study [20]. These systems share the same privacy problem. Complete transparency of donor-recipient relationships exposes identities and amounts to anyone with an Ethereum node. A growing research trend in Muslim-majority countries was confirmed by Fauzi et al. [19] through bibliometric analysis. Islamic social finance instruments (zakat, infaq, sadaqah, waqf) were shown by Ascarya [20] to support economic recovery, while Unal and Aysan [21] proposed blockchain as a way to harmonize differing Sharia-compliance regimes. The blockchain-Islamic social finance nexus for sustainability was explored by Muharam and Osman [26]. None integrate privacy-preserving transaction mechanisms.

The ZKP construction space has been surveyed extensively by Hameed et al. [7], while zk-SNARK variants for blockchain were analyzed by Wen et al. [8]. Williams [21] situates ZKPs within the blockchain landscape. Poseidon, introduced by Grassi et al. [11], cut constraint counts by up to 8× compared to Pedersen Hash on EVM chains. This work uses Noir [10], a Rust-like DSL compiling to ACIR, with Barretenberg's UltraHONK prover generating the proof and a Solidity verifier contract checking it on Sepolia. The pipeline stays client-side, meaning private inputs never leave the donor's device. Proof hashes anchor on-chain for auditability. A ZKP-based framework for privacy-preserving, regulation-compliant transactions was proposed by Buterin et al. [15], and zk-SNARK identity management for KYC integration was presented by Mühle et al. [18].

Three gaps stand out. Existing blockchain zakat platforms, as Nazeri et al. [3] acknowledge with "complete traceability of zakat funds," lack any privacy-preserving transaction mechanism. General-purpose ZKP frameworks [15, 14] provide anonymity without Islamic eligibility predicates. They can hide a transaction but cannot prove the sender meets nisab or hawl. No prior work combines programmable privacy (Noir circuits, verifiable predicates) with Islamic social finance instruments [21, 5]. This paper targets these gaps.

Fig. 1 positions this research relative to three adjacent domains and the Design Science Research methodology. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14,15]. Islamic social finance studies digitize zakat without ZK integration [20,21,24,25]. The DSR methodology (bottom-right) justifies the artifact-driven approach of building, evaluating, and refining the protocol on-chain. ZKT bridges these domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification, instantiated and validated via Sepolia deployment (v8).

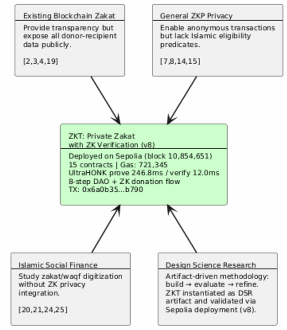

Fig. 1. Research positioning of ZKT relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZKT model that preserves privacy through off-chain proving and bounded on-chain disclosure.

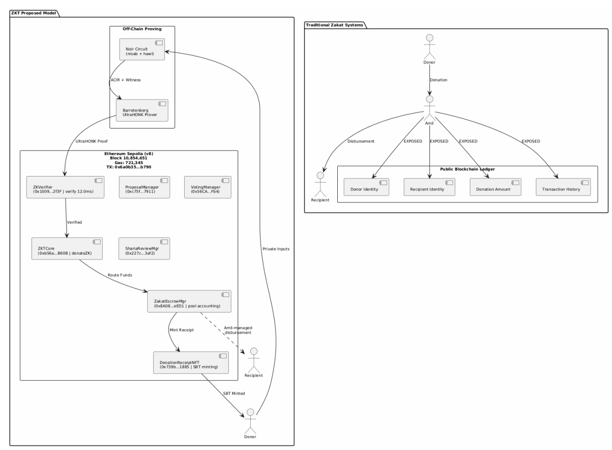

Fig. 2. Traditional Zakat Paradigm vs. Proposed ZKT Model

III. Methodology

This study uses a Design Science Research (DSR) approach, mapped across five phases in Fig. 3. The search began with 29 papers spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments published between 2021 and 2025 (Phase 1). Protocol design followed (Phase 2), covering Noir circuits for nisab and hawl and the four-tier privacy model. Phase 3 built the system with nargo v1.0.0-beta.21, Barretenberg v4.2.1, 15 Solidity contracts deployed via a single forge script to Sepolia, and a Next.js 15 PWA frontend with wagmi/XellarKit wallet connectivity. The prototype then ran on Sepolia (Phase 4), producing proofs in 247 ms and verifying them in 12.0 ms. Phase 5 evaluated with Foundry gas reports, security properties analysis, and Sharia compliance review.

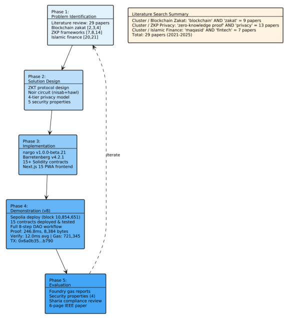

Fig. 3. DSR Methodology with five phases and literature search summary

IV. Results

A. System Overview and Threat Model

The ZKT protocol uses a five-actor trust model. The actors are donor (ZKP prover), recipient (SBT receipt holder), amil institution (verifier), off-chain proving infrastructure (Noir circuits and Barretenberg prover), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a potentially malicious claimant, and (3) a public blockchain network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

TABLE I
ZKT Protocol Security Properties

Property	Guarantee
Donor anonymity	Identities and amounts never exposed on-chain. Only ZK proofs are publicly verifiable.
Recipient privacy	Donor-side privacy via ZK proofs. Recipient encrypted notes reserved for future work.
Soundness	Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].
Double-donation	Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.
Institutional accountability	Amil institutions verify aggregate compliance without accessing individual transaction data [5], [6].

Fig. 4 shows the three-layer protocol architecture. The off-chain proving pipeline compiles Noir circuits to ACIR and generates UltraHONK proofs via Barretenberg at 247 ms. The Sepolia settlement layer hosts ZKTCore and ZakatEscrowManager. The Next.js PWA frontend uses wagmi/XellarKit for wallet connectivity.

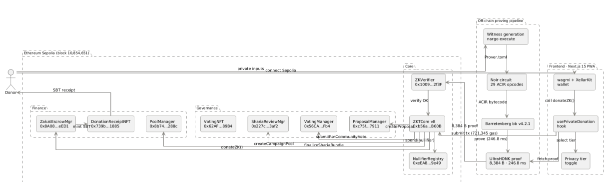

Fig. 4. ZKT Protocol Architecture: off-chain proving pipeline and on-chain settlement

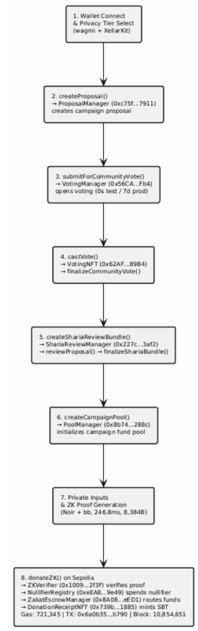

Fig. 5. ZKT Donation Flow: from wallet connection to SBT receipt to the donor

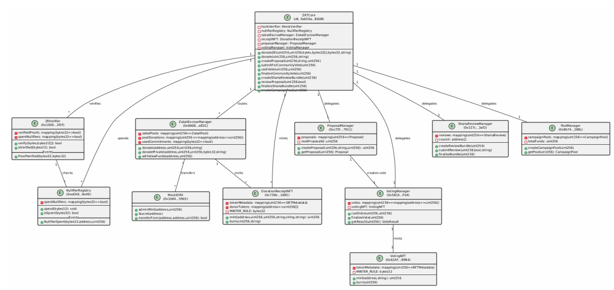

Fig. 6. ZKT Smart Contract Architecture with Sepolia addresses and cardinality

B. System Architecture and Payment Flow

Noir circuits run client-side. Ethereum Sepolia handles settlement, donors choose stablecoins or ETH, and everything flows through `donateZK()`. Proof generation happens off-chain. Verification on-chain. Privacy comes from the proofs, accountability from the events and nullifiers.

The donor goes through a Progressive Web Application, connecting a wallet, selecting a privacy tier, entering an amount, watching proof generation, and confirming the transaction. A Web Worker runs the actual proving, so private inputs never leave the donor's device [11].

C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v4.2.1 on a 16-core Linux machine. Gas measurements for standard contract functions were obtained from Foundry gas reports. The end-to-end donateZK() gas was measured on the Sepolia testnet at block 10,854,651.

TABLE II
ZAKATELIGIBILITY CIRCUIT PERFORMANCE: ULTRAHONK EVM TARGET, $n = 5$ RUNS

Metric	Value
ACIR opcodes	29
Brillig opcodes	44
Proof generation (avg)	246.8 ms
Proof verification (avg)	12.0 ms
Proof size	8,384 bytes

D. Sepolia Deployment and Gas Consumption

15 contracts deployed via single forge script to Sepolia (chain ID 11155111). The ZKTCore (v8, 0xb56a...B60B) routes `donateZK()` through `zakatEscrowManager.donate()` for pool accounting. An end-to-end donation (tx 0x6a0b...b790, block 10,854,651) completed proof verification, nullifier registration, transfer, NFT minting, and event emission in 721,345 gas.

Table III reports gas costs measured via Foundry gas reports.

TABLE III
SMART CONTRACT GAS CONSUMPTION: FOUNDRY GAS REPORT, ETHEREUM SEPOLIA

Contract	Function	Gas (avg)
ZKTCore	`donate()`	490,187
ZKTCore	`createCampaignPool()`	365,552
ZKTCore	`castVote()`	132,909
ZKTCore	`donateZK()` (e2e)	721,345
ZKTCore	Deploy	5,314,471
ZakatEscrowMgr	Deploy	3,069,194
ShariaReviewMgr	Deploy	2,114,319

E. Security Analysis

Donor anonymity. The only on-chain observable from a private donation is the event emitted by `ZKTCore.sol` (lines 577-582):

```
event ZKDonationReceived(
    uint256 indexed poolId,
    bytes32 indexed nullifier,
    uint256 amount,
    address indexed donor
);
```

Private mode zeroes both `donor` and `amount`, nothing traceable, while Semi-Private attaches the donor's address but hides the amount as a design tradeoff for institutional accountability. Neither mode leaks the full `(donor, amount)` pair. The verifier contract checks only a hash and a nullifier flag. That is the entire on-chain footprint.

Recipient privacy. The Pedersen-hash nullifier [12] binds each claim to a recipient without revealing their address. The nullifier computation $\text{pedersen}(secret, recipient\_address, cycle\_id)$ in `main.nr` (lines 35-39) means the on-chain nullifier is a deterministic but preimage-resistant value. An observer sees `0x3f2a...` but cannot recover the recipient address or secret. Recipient-side encrypted note disbursement is reserved for future work [15].

Circuit soundness. Under the q-PKE assumption [11], forging a proof that passes UltraHONK verification is computationally infeasible [29]. A non-eligible recipient (one whose wealth exceeds the nisab threshold, as checked by `verify_nisab()` at line 7-11 of `main.nr`) cannot produce a witness that satisfies the circuit constraints.

Double-donation prevention. The on-chain `NullifierRegistry` (line 10 of `NullifierRegistry.sol`) maintains a `mapping(bytes32 => bool)` of spent nullifiers. The `spend()` function (line 19-23) enforces single-use: `require(!spentNullifiers[nullifier])` rejects any nullifier already recorded. The `ZKTCore.donateZK()` function (line 553-572) atomically chains `honkVerifier.verify()` → `nullifierRegistry.spend()` → `zakatEscrowManager.donate()`, so a duplicate nullifier causes the entire transaction to revert. Each eligible recipient can claim zakat at most once per cycle. Although the nullifier hash is preimage-resistant (an observer cannot derive the recipient address from it), the address itself is a public circuit input, visible in the transaction calldata. Full recipient privacy, where not even the amil institution can identify claimants, depends on encrypted note disbursement, reserved for future work.

V. DISCUSSION

A. Sharia Compliance and Deployment

Five *daruriyyat* anchor maqasid al-shariah. Religion, life, intellect, progeny, and wealth [5]. Al-Shatibi wrote this centuries ago. Ibn Ashur later stretched *hifz al-nafs* to cover dignity, arguing that "exposing an individual's financial weakness to the public constitutes harm to their honour" [6]. Whether he would have imagined blockchain is doubtful, but the principle maps onto it directly. A single address recorded as a zakat recipient on Etherscan broadcasts below-nisab status permanently.

Azizov et al. [25] propose a maqasid al-shariah framework for fintech regulation. Latang et al. [24] examine the Islamic legal status of cryptocurrency assets. Pramuka et al. [27] explore blockchain's role in Islamic charitable crowdfunding. This work addresses the donor side through ZK proofs and nullifier-based anonymity. Recipient-side privacy through encrypted private notes is reserved for future work.

Two maqasid dimensions are in play. *Hifz al-nafs* appears through Private and Semi-Private modes, which keep donations and linkages off-chain. *Hifz al-mal* is served by the nullifier registry stopping double-claims and the proof's soundness blocking fake eligibility. *Amanah*, institutional trust, flows from the `ProofVerified` event, which lets amil institutions audit aggregate flows without touching individual records.

Beyond Sepolia, three pieces must fall into place. First, Indonesia's OJK Fintech Sandbox approval, training for amil institution staff on wallet management and proof verification, and a real rupiah-backed digital currency to replace the mock IDRX stablecoin.

B. Limitations and Future Work

The circuit covers nisab and hawl. Eight asnaf categories are architecturally possible. The amil institution disburses through ZakatEscrowManager today with no on-chain proof the money reached the mustahik. DIDs for recipients would close that gap and make disbursements verifiable. A dual governance model, Sharia Council ZK DAO plus Community Donor DAO, is under design. Other directions: recursive proof systems for multi-period eligibility, compliance-friendly ZK protocols for AML/CFT, and batched settlement optimization [7], [17], [23].

VI. Conclusion

Using Noir circuits and UltraHONK proofs, we deployed a privacy-preserving zakat protocol on Ethereum Sepolia that shows accountability and privacy need not fight each other. The circuit, 29 ACIR opcodes, verifies nisab and hawl client-side, a Pedersen-hash nullifier stops double-claims, and a four-tier model gives donors control over disclosure. Five benchmark runs averaged 246.8 ms (Barretenberg v4.2.1). One end-to-end Sepolia transaction cost 721,345 gas.

It turns out privacy and Sharia compliance pull in the same direction, not opposite ones. Ibn Ashur made dignity central to nafs protection [6], and ZK proofs make that operational, verifying eligibility without exposure. The `ZKDonationReceived` event gives amil institutions an auditable aggregate record while the nullifier registry prevents abuse. What the blockchain shows and what it reveals about individuals need not overlap.

Future work extends the circuit to all eight asnaf categories, adds recipient-side encrypted note disbursement, and pursues OJK Fintech Sandbox integration for production deployment.

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